```
        panel.grid.major = element_blank()) +
  labs(title = "Spatial Prisoner's Dilemma: Grid Snapshots",
       x = NULL, y = NULL)

p_snapshots
save_pub_fig(p_snapshots, "spatial-pd-snapshots", width = 8, height = 8)
```

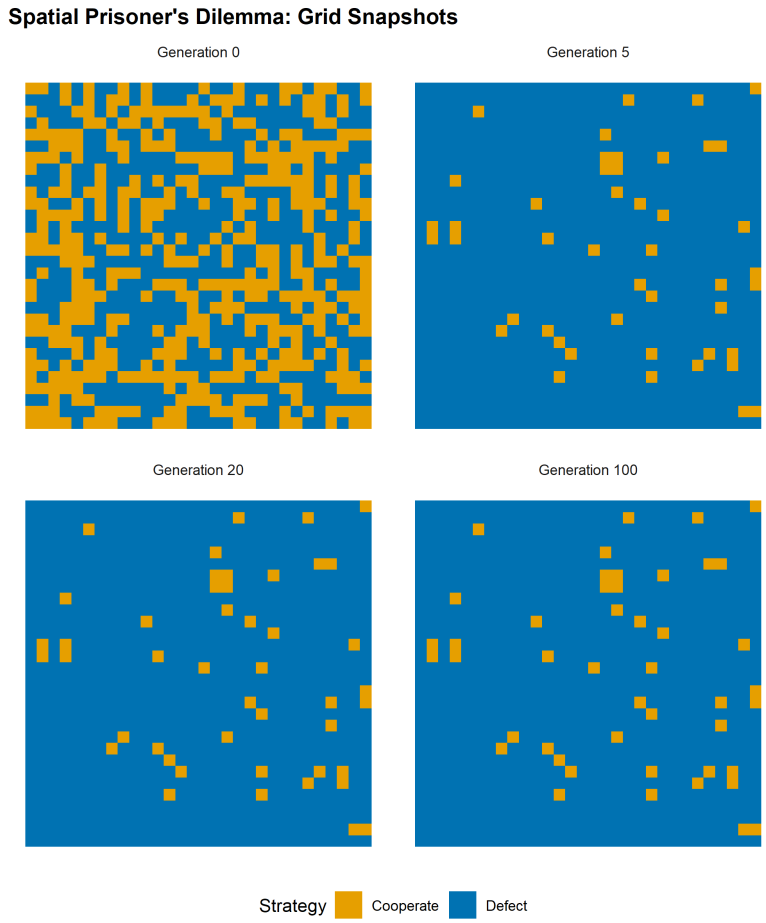

Figure 39.1.: Grid snapshots at generations 0, 5, 20, and 100. Cooperators (orange) form compact clusters that resist invasion by defectors (blue). By generation 20, the spatial pattern has largely stabilized into a dynamic equilibrium.

The initial random arrangement (generation 0) shows a salt-and-pepper mixture. By generation 5, cooperators have begun to aggregate into clusters while isolated cooperators surrounded by defectors have been converted. By generation 20, the pattern has largely stabilized: compact cooperator clusters persist, separated by defector boundaries. The clusters are not static — their edges shift over time — but the overall cooperation level remains roughly stable.

```
coop_df <- tibble(
  generation = 0:100,
  cooperation = result$coop_frac
)
```

```r
p_coop <- ggplot(coop_df, aes(x = generation, y = cooperation)) +
  geom_line(colour = okabe_ito[1], linewidth = 1) +
  geom_hline(yintercept = mean(result$coop_frac[51:101]),
             linetype = "dashed", colour = okabe_ito[8]) +
  scale_y_continuous(labels = label_percent(), limits = c(0, 1)) +
  theme_publication() +
  labs(title = "Cooperation Fraction Over Time",
       x = "Generation", y = "Fraction cooperating")

p_coop
save_pub_fig(p_coop, "spatial-pd-cooperation", width = 7, height = 5)
```

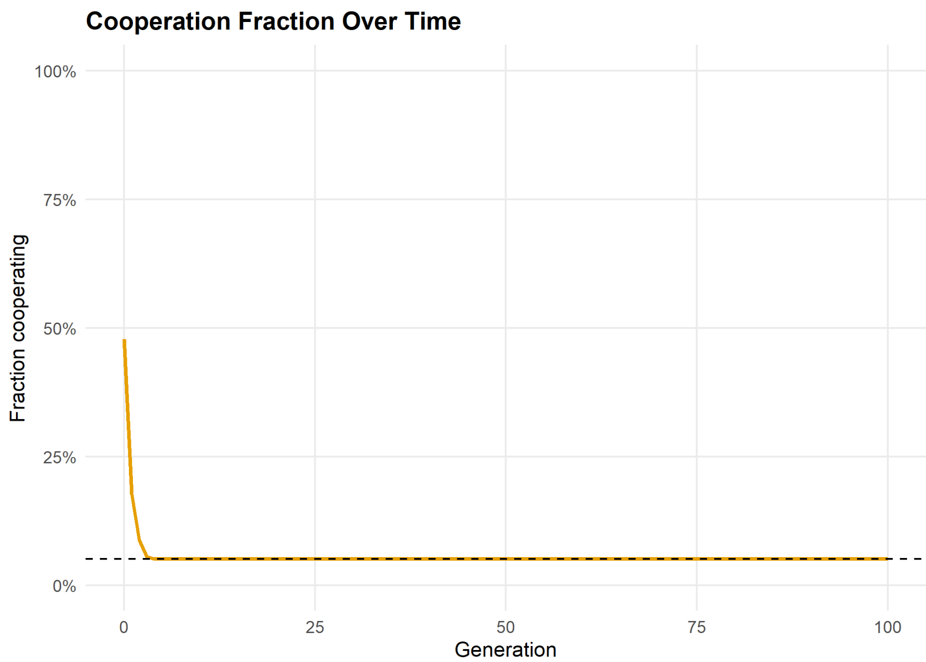

Figure 39.2.: Cooperation fraction over 100 generations. After an initial transient where isolated cooperators are eliminated, the cooperation rate stabilizes around a dynamic equilibrium. Spatial structure sustains a substantial level of cooperation even in the one-shot Prisoner's Dilemma.

```r
cat(sprintf("Initial cooperation rate: %.1f%%\n",
            result$coop_frac[1] * 100))
```

```
#> Initial cooperation rate: 47.8%
```

```r
cat(sprintf("Final cooperation rate:   %.1f%%\n",
            result$coop_frac[101] * 100))
```

```
#> Final cooperation rate:   5.1%
```

```r
cat(sprintf("Mean cooperation (gen 50-100): %.1f%%\n",
            mean(result$coop_frac[51:101]) * 100))
```

```
#> Mean cooperation (gen 50-100): 5.1%
```

The cooperation rate drops sharply in the first few generations as isolated cooperators are eliminated. It then stabilizes as surviving cooperators form self-reinforcing clusters. The long-run cooperation rate depends on the payoff parameters — with the canonical values used here, spatial structure sustains cooperation at a level far above zero, even though defection would dominate in a well-mixed population.

39.5. Extensions

- **Varying temptation**: As the temptation payoff T increases (holding other payoffs fixed), the sustainable cooperation rate decreases. There is a critical threshold above which cooperators go extinct even with spatial structure. Experimenting with this threshold is a good exercise.
- **Stochastic update rules**: Instead of deterministic imitation of the best neighbor, one can use probabilistic rules where the probability of adopting a neighbor's strategy increases with the neighbor's payoff advantage. This connects to the **Fermi update rule** and reduces sensitivity to small payoff differences.
- **Network structure**: The lattice is a special case of a network. Cooperation dynamics on random graphs, scale-free networks, and small-world networks exhibit qualitatively different behavior (Chapter 45).
- **Evolutionary stability**: The spatial model shows that cooperation can be an evolutionarily stable configuration even when it is not an ESS in the well-mixed case (Chapter 43).
- For the original model, see Nowak and May (1992). For a comprehensive review of spatial evolutionary games, see Szabó and Fáth (2007).

Exercises

1. **Effect of grid size.** Run the spatial PD on grids of size 10×10, 30×30, and 50×50 with the same initial cooperation probability (50%). How does the long-run cooperation rate depend on grid size? Explain why boundary effects matter more on smaller grids.

2. **Temptation threshold.** Fix $R = 3$, $P = 1$, $S = 0$ and vary T from 3.5 to 8 in increments of 0.5. For each value, run the simulation for 100 generations and record the final cooperation rate. Plot cooperation rate vs. T and identify the approximate threshold where cooperators go extinct. (*Hint*: modify `play_round()` to accept custom payoffs.)

3. **Initial cluster.** Instead of random initial placement, start with a single 5×5 block of cooperators in the center of a 30×30 grid of defectors. Does the cooperator block survive? Grow? How does the outcome depend on T?

Solutions appear in Appendix H.